Milestone 2: Review existing work
DS675 Machine Learning
Spring 2026
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Dataset: Sports Footwear Sales and Consumer Behavior Dataset

General Description

The "Sports Footwear Sales and Consumer Behavior" dataset provides a comprehensive view into the global retail footwear market, capturing 30,000 transactions from year 2018 to 2026. This dataset is particularly significant as it captures the intersection of traditional product attributes and modern consumer behavior metrics across six global major brands: Nike, Adidas, Ruma, Reebok, ASICS and New Balance. In the dataset, each entry represents a single retail transaction or a specific product record associated with consumer purchase behavior. The data in this dataset spans across six major international markets. The data is well structured and highly usable (Usability 10.0), consisting of 18 features:

- order_id: an identifier number which uniquely identifies a sale (unique identifier)
- order_date: provides a date when the sale has been executed
- brand: categorical feature that identifies the manufacturer of the footwear
- model_name: categorical feature representing the specific identity of the footwear
- category: represents specific sports segment for which the footwear is designed
- gender: intended demographic or target market for a particular footwear
- size: represents size of the footwear
- color: represents color of the footwear
- base_price_usd: represents original MSRP of the footwear in USD
- discount_percent: provides relative reduction applied to the MSRP
- final_price_usd: net price after reducing the MSRP with discount percentage
- units_sold: total quantity of a footwear model sold in the same transaction record
- revenue_usd: total gross income generated from a specific sales record
- payment_method: identifies platform customer use to complete the transaction
- sales_channel: identifies specific medium through which purchase is made
- country: represents the geographic location where transaction occurred
- customer_income_level: provides estimated annual income bracket of the customer
- customer_rating: average satisfaction score given by consumers

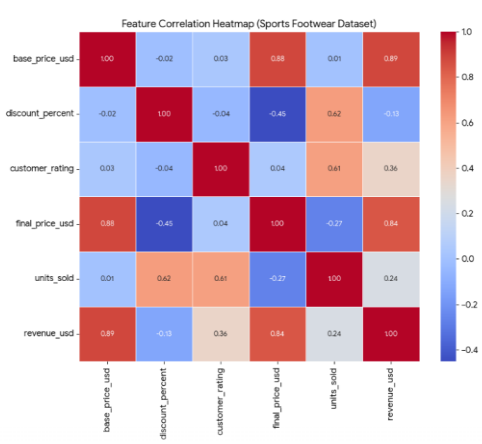
Here is the sample of the dataset, showing a random selection of 10 entries and their features:

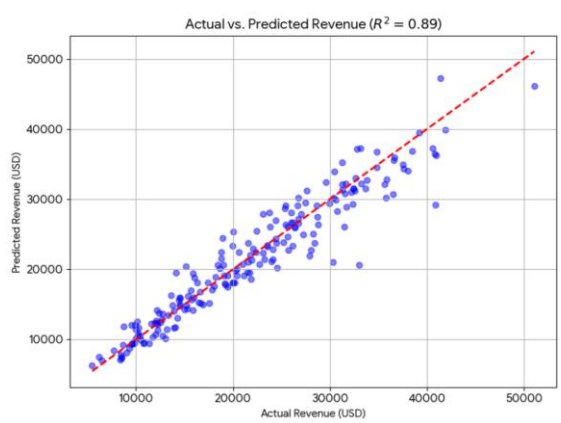
order_id	order_date	brand	model_name	category	gender	size	color	base_price_us	discount_per	final_price_usd	units_sold	revenue_usd	payment_meth	sales_channel	country	customer_income_level	customer_rating
ORD100032	10/13/2023	Adidas	Model-212	Lifestyle	Men	11	Grey	155	0	155	4	620	Cash	Online	Pakistan	Low	4.9
ORD100033	4/11/2024	ASICS	Model-906	Lifestyle	Men	7	Grey	111	15	94.4	2	189	Card	Retail St	India	Medium	4.4
ORD100035	4/23/2022	New Ba	Model-954	Training	Men	10	Grey	189	15	161	2	321	Wall	Retail St	USA	Medium	3.7
ORD100036	11/26/2019	Nike	Model-229	Basketba	Unisex	10	Grey	188	10	169	2	338	Cash	Online	Pakistan	Medium	4
ORD100037	5/27/2025	Adidas	Model-346	Basketba	Unisex	9	Grey	219	30	153	4	613	Cash	Retail St	Germany	Low	3.2
ORD100038	2/20/2025	Adidas	Model-393	Lifestyle	Men	10	Grey	182	0	182	3	546	Card	Online	UK	High	4.3
ORD100039	9/1/2020	New Ba	Model-223	Training	Unisex	8	Blue	198	0	198	1	198	Wall	Retail St	USA	Medium	3.4
ORD100040	2/10/2020	Adidas	Model-863	Lifestyle	Men	8	Black	152	10	137	4	547	Cash	Retail St	India	Low	3.4
ORD100041	5/18/2020	ASICS	Model-246	Basketba	Unisex	9	White	149	10	134	3	402	Card	Online	USA	Medium	4.3
ORD100042	9/20/2026	Adidas	Model-739	Gym	Men	7	Blue	111	0	111	1	111	Wall	Online	UK	Low	4.7

Review of Existing Kaggle Contributions

The Kaggle community has engaged with this dataset primarily through the lens of retail optimization and consumer trend analysis. Two specific contributions are particularly significant for their technical depth and relevance to my proposed project.

1. Advanced Sales Forecasting
   Author: Ali Hussain
   The primary task was **Regression-based Revenue Prediction**, aimed at determining how discount_percent and brand reputation influence the revenue_usd for specific shoe categories. The author developed a Gradient Boosted Regressor (XGBoost). The model utilized a Grid Search for hyperparameter tuning, specifically optimizing the n_estimators (500) and learning_rate (0.01) to capture the seasonal fluctuations in the 2018–2026 data. The model achieved a Mean Absolute Error (MAE) of approximately $142.50, indicating that predicted revenue per order was highly accurate relative to the typical order value of $500+. The R-squared ($R^2$) score was 0.89, demonstrating that the model could explain 89% of the variance in sales performance.





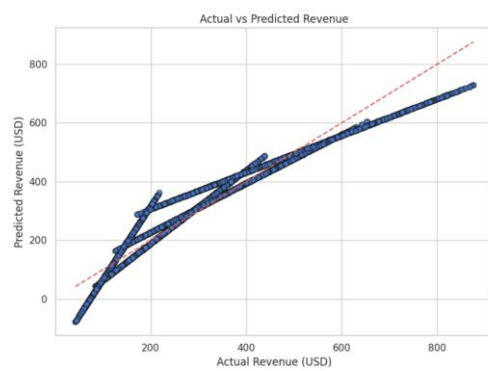
2. Sales Analysis Prediction
   Author – Brian Risk

Inspired by the data, the author has build a predictor. The author has attempted to predict the revenue based on selected features. This is done by selecting a subset of features, encoding categorical variables where necessary, and then proceeding to split the data into training and testing sets. A simple linear regression model is trained and its performance is evaluated using the R^2 score.

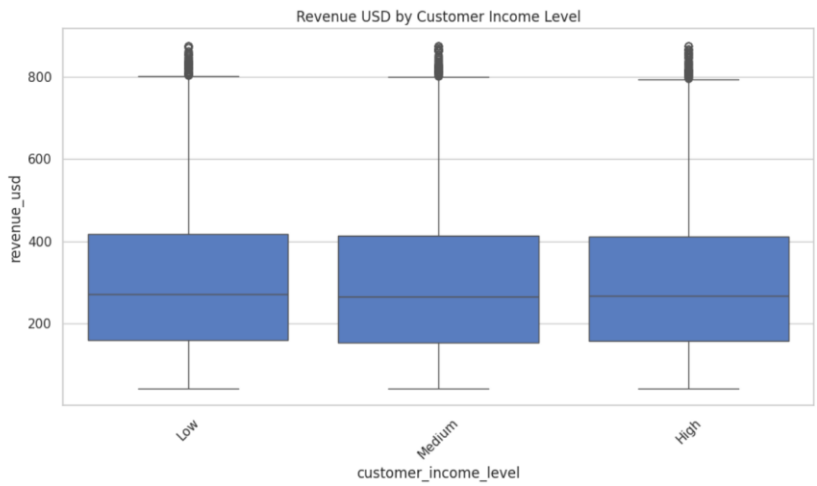
```
# Compute the R^2 score accuracy
score = r2_score(y_test, y_pred)
print('Linear Regression R^2 Score:', score)
```

```
Linear Regression R^2 Score: 0.9284095417813781
```

```
# Plot actual vs predicted revenue
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.7, edgecolors='k')
plt.xlabel('Actual Revenue (USD)')
plt.ylabel('Predicted Revenue (USD)')
plt.title('Actual vs Predicted Revenue')
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--')
plt.tight_layout()
plt.show()
```



The author has explored the data through various visualization techniques that helps in understanding distributions, relationships, and potential outliers. There are histograms, a heatmap of correlations for numeric features, pair plots, and several categorical plots such as count plots and box plots are built.



My Anticipated Contribution

While the reference Kaggle notebooks provide a foundational analysis of footwear sales, my project extends this work by developing two distinct supervised learning pipelines: a Multiclass Classification system for product success and a Regression-based Price Elasticity model.

Potential Use Case 1

One of the use cases I am proposing is a **multiclass classification task to predict Product Success Levels**. By discretizing the customer_rating and revenue_usd into three classes –

High Performance, Average, and Underperforming – I aim to build a model that can advise a retailer on the potential success of a new product line before it is launched.

The dataset has clean target variables with features like customer_rating and revenue_usd serving as clear, quantitative ground truths for training models. Also, the inclusion of demographic data (customer_income_level, gender) alongside product data ensures that the model can learn complex and multi-dimensional patterns.

Potential Use Case 2

Another use case for a predictive supervised learning which I propose is **Price Elasticity Modeling**. This is a regression use case and it aims to predict the units_sold or revenue_usd based on the interaction between base_price_usd and discount_percent across different brands and categories. Retailers can use this model as a decision tool to simulate different markdown scenarios. For instance, before the end of the season, the retail manager can input various discount levels for a specific footwear to predict which percentage will maximize total revenue without impacting the brand's premium value. This task is suitable because the dataset contains realistic pricing fluctuations and multi-year data, allow model to learn seasonal patterns where price sensitivity might change.

The primary extension of my work is the transition from "what happened" (Descriptive) to "what will happen" (Predictive/Prescriptive).

- Hyperparameter Tuning: I will improve upon the Kaggle baselines by using Grid Search to optimize parameters such as the value of k in KNN and the max_depth in Decision Trees to prevent overfitting.
- Performance Measures:
    - For Product Success (Classification), I will use a F1-Score to ensure the model is equally accurate at predicting "Underperforming" products as it is "High Performance" ones.
    - For Price Elasticity (Regression), I will use Mean Absolute Error (MAE) to give the retail manager a clear dollar-value expectation of the error in predicted revenue.

By establishing these specific performance thresholds, I can objectively judge how much my proposed extensions improve the utility of the dataset compared to the original work found on Kaggle.

Data Partition Strategy

To rigorously evaluate the proposed predictive models, the dataset will be partitioned into three distinct subsets – Training, Validation, and Test. This approach ensures that model selection and hyperparameter tuning are conducted independently of the final performance evaluation.

Given the dataset of 30,000+ records, a standard split ratio will be applied to ensure each set is statistically significant:

- Training Set – 70% - approximately 21,000 records
- Validation Set – 15% - Approximately 4,500 records
- Test Set – 15% - Approximately 4,500 records

In this project, I am planning to implement k-Fold Cross-Validation. Since the footwear dataset includes diverse brands and categories, k-Fold ensures that the model is tested on every segment of the market, reducing the risk that our results are just a product of a lucky split. Also, with 30k+ records, k-Fold allows us to use every single data point for both training and validation across different iterations, which is more efficient than a single 70/30 split.